

Diagnosis information - Airbag

Airbag control unit Plug contacts - Electrical fault

⚠ WARNING

Uncontrolled triggering of the airbag

- Contusions, crushing injuries or death

- ▶ **Never hang the PIWIS Tester** on or over the steering wheel.
- ▶ Before working on the ignition circuits of the airbag system, always disconnect the battery from the vehicle electrical system and ensure that the vehicle electrical system is de-energized.
- ▶ Wait for at least one minute before starting work on the airbag system.
- ▶ Perform fault finding on the ignition circuits of the airbag system and its lines only using approved special tools.

⚠ WARNING

Driving with attached/connected special tools or commercially available tools

- Contusions, crushing injuries or death
- Traffic hazard

- ▶ Never drive with special tools and/ or commercially available tools attached instead of the airbag units.

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on
- Airbag triggering unit Internal function test

Possible fault causes

❏ This fault is set even though the corresponding component is not installed in the vehicle. However, the contact pins for these components are present on the control unit. If moisture now penetrates into the connector, for example, this can produce short circuits and cause electrical faults to be set.

- ♦ Short circuits at the pins of the control unit/within the connector housing of the control unit, e.g. due to moisture

Fault effects

- No functional restrictions

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Airbag sensor Lock carrier Driver

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- ♦ **Coding** - Ignition circuit is connected and OK, but is not coded.
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- ♦ **Airbag sensor** incorrect
- ♦ **Airbag sensor** Faulty

Supply voltage with disconnected plug Airbag sensor and connected Plug control module Airbag

- Ignition on
- Measurement/s at the plug contacts **Airbag sensor** must be performed (Plug disconnected)
- Nominal value : 2.0 V $\Rightarrow$ 2.5 V (measured value fluctuates by +/- 0.15 V)

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Short circuit to B+:

- ♦ Setpoint voltage **- 0.5 V $\Rightarrow$ + 0.5 V**

Short circuit to ground:

- ♦ Setpoint resistance **> 0.9 MOhm**

Open circuit:

- ♦ Setpoint resistance **< 2 Ohm**

Short circuit between lines :

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- ♦ Warning light for Airbag is **ON**
- ♦ **Control module** Airbag Limited function / No function

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- Ignition off > 2 Second(s), then:
- Ignition on > 5 Second(s)
- Airbag triggering unit Internal function test

Possible fault causes

- ♦ **Coding** - Ignition circuit is connected and OK, but is not coded.
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- ♦ Short circuit to B+/ground/between lines or open circuit in the lines
- ♦ **Airbag sensor** incorrect
- ♦ **Airbag sensor** Faulty

Supply voltage with disconnected plug Airbag sensor and connected Plug control module Airbag

- Ignition on
- Measurement/s at the plug contacts **Airbag sensor** must be performed (Plug disconnected)
- Nominal value : 2.0 V $\Rightarrow$ 2.5 V (measured value fluctuates by +/- 0.15 V)

Supply voltage with connected plug Airbag sensor and connected Plug control module Airbag

- Ignition on
- Measurement/s at the plug contacts **Airbag sensor**
- Nominal value : 8.0 V $\Rightarrow$ 8.2 V

NOTE

The specified values and value ranges correspond to the OK state in each case.

Line test without connected components

Short circuit to B+:

- ♦ Setpoint voltage **- 0.5 V $\Rightarrow$ + 0.5 V**

Short circuit to ground:

- ♦ Setpoint resistance **> 0.9 MOhm**

Open circuit:

- ♦ Setpoint resistance **< 2 Ohm**

Short circuit between lines :

- ♦ Setpoint resistance **> 0.9 MOhm**

Fault effects

- ♦ Warning light for Airbag is **ON**
- ♦ **Control module** Airbag Limited function / No function

Diagnosis information - Airbag

Airbag sensor Lock carrier Passenger

⚠ WARNING

Uncontrolled triggering of the airbag

- **Contusions, crushing injuries or death**

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- ▶ Perform fault finding on the ignition circuits of the airbag system and its lines only using approved special tools.

⚠ WARNING

Driving with attached/connected special tools or commercially available tools

- **Contusions, crushing injuries or death**
- **Traffic hazard**

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Diagnostic conditions

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- ♦ Warning light for Airbag is **ON**
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Diagnosis information - Airbag

Airbag sensor Lock carrier Passenger

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Diagnosis information - Airbag

Airbag sensor Lock carrier Passenger

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Fault effects

- ♦ Warning light for Airbag is **ON**
- ♦ **Control module** Airbag Limited function / No function

Diagnosis information - Airbag

Airbag sensor Passenger's side front

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Uncontrolled triggering of the airbag

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Diagnosis information - Airbag

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